Solve — Crafting the *Lapis Philosophorum*

the *Automatica Principia Proceduralis*

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The Philosopher's Stone emerges as a 12-primitive crystal via the 4 classical alchemical operations — *Separatio*, *Purificatio*, *Albedo*, & *Rubedo*.

0 – *Prima Materia.* Performing the operations on ordinary, inert matter was an understandable, categorical error of history's alchemists.

However some, like Luria, knew *precisely* the material upon which the *Work* was to be done.

Take: abstract category $\mathcal{C}$.

No primitives yet — the starting object has no invariants beyond the axioms of a category itself.

In alchemical terms, this was always the intended *prima materia*: the undifferentiated substance from which the *Work* begins.

1 – *Separatio* — Logical on $\mathcal{C}$.

- L1 (adjunction) $\to R$ (relational mode: adjoint structure)
- L2 (dagger) $\to R$ (confirms and refines R value)

R is the first primitive to crystallize: it is a logical property of the morphisms of $\mathcal{C}$ directly.

This is the *separatio*: the first extraction of form from the undifferentiated.

2 – *Purificatio* — Inductive on $\mathcal{C}$.

- I2 (iteration) $\to H$ (temporal depth: depth of stabilization)
- I3 (classifying) $\to \Omega$ (winding: π_1 of $B\mathcal{C}$)
- I4 (Yoneda) $\to D$ (dimensionality: $D_\odot$ if Yoneda is point-surjective; free cocompletion size gives D_∞, $D_\triangle$, $D_\wedge$)

I4 also introduces the possibility of asking L4 (Lawvere) later — you cannot ask whether $A \to A^A$ is surjective until A^A exists, which requires cartesian closure from Stage 3.

This is the *purificatio*: the base substance is washed, dissolved, and reconstituted at higher resolution.

3 – *Albedo* — Algebraic on $\mathcal{C}$.

- A1 (monoidal) $\to S$ (stoichiometry: arity of generators); P (parity: symmetry of tensor)
- A2 (monad) $\to K$ (kinetics: spectral properties of μ)
- A3 (dagger) $\to F$ (fidelity: unitality of duals)
- A4 (enriched) $\to \Gamma$ (interaction grammar: enrichment base)

After Stage 3, $\mathcal{C}$ is a symmetric monoidal dagger category with a monad and enrichment.

All shape primitives are determined.

The gate primitives (T, Φ) require one more step.

This is the *albedo*: the whitening, the full structuring — all intermediate forms are now in place.

4 – *Rubedo* — Logical on algebraically-enriched $\mathcal{C}$.

- L3 (cartesian closure) $\to T$ (topology: does the state space admit internal homs?)
- L6 (paraconsistent negation) $\to$ determines the truth-value structure in which L4 will be evaluated
- L4 (Lawvere) under L6 $\to \Phi$:
  - Classical (L4 holds, no L6): Φ_c
  - Dialetheic (L4 holds via LP truth value $\mathbf{b}$): $\Phi_c^{\mathbb{C}}$
  - Inclosure (L4 form holds, content contradicts via Inclosure Schema, no explosion because L6 present): Φ_{EP}
  - L4 fails entirely: Φ_{sub} or Φ_{super}
- L5 (Frobenius) $\to P$ promotes to $P_{\pm}^{\mathrm{sym}}$ if $\mu \circ \delta = \mathrm{id}$

G (scope) emerges from *Albedo* + *Purificatio*: the correlation length of the Yoneda embedding under the monoidal structure.

Local ($G_{\beth}$) if the tensor decouples; global ($G_{\aleph}$) if the Yoneda embedding is faithful across all scales.

This is the *rubedo*: the reddening, the final perfection.

The gate primitives lock.

The Stone is complete.

Sic transit gloria categoriæ.